APPARATUS FOR A MASSIVELY PARALLEL OPTICAL DATA BUS ARCHITECTURE

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Title: APPARATUS FOR A MASSIVELY PARALLEL OPTICAL DATA BUS ARCHITECTURE

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Cross-Reference to Related Applications

This application claims the benefit of and priority to U.S. Patent Application No. 19/559,705, the contents of which are hereby incorporated by reference in their entirety.

Abstract

An apparatus for a computer data bus with a data path width greater than 128 bits, implemented via a plurality of parallel optical channels. The apparatus overcomes the physical limitations of electrical buses, such as signal crosstalk, timing skew, and pin density, which have made architectures wider than 64 bits impractical. By grouping a small number of high-cardinality optical channels (e.g., pixels) to operate in concert, the invention enables the creation of ultra-wide data buses (e.g., 256-bit, 1024-bit) in a microscopic footprint, providing a foundational component for next-generation, high-throughput computing architectures.

Background of the Invention

For decades, computer bus architectures have been constrained by the physical limitations of parallel electrical signaling. Increasing the width of a data bus from 32 to 64 bits was a significant engineering challenge, requiring more physical pins on CPU packages and complex trace routing on motherboards. Architectures wider than 64 bits, such as 128-bit, have been largely impractical for general-purpose computing due to insurmountable problems with signal integrity, timing skew between the parallel lines, and the physical space required for the electrical connections. This "width wall" has forced designers to increase clock frequency to achieve higher throughput, leading to greater power consumption and heat generation.

Summary of the Invention

The present invention is an apparatus for a massively parallel optical data bus that shatters the conventional width limitations of electrical buses. The apparatus uses a small plurality of parallel optical channels, such as the pixels of an optical interface, to form a single, ultra-wide data bus.

The core of the apparatus is a data bus where a data word is transmitted in a single cycle across multiple optical channels operating in concert. For example, a 128-bit data word can be transmitted by grouping six 24-bit optical channels (pixels), providing an effective bus width of 144 bits. A 1024-bit data bus can be realized with as few as 43 optical channels. This is achieved in a physical footprint that is orders of magnitude smaller than would be required for an equivalent electrical bus, and without the associated signal integrity problems. This apparatus makes 256-bit, 512-bit, 1024-bit, and wider bus architectures practical for the first time, enabling a paradigm shift in computer design from frequency scaling to parallelism scaling.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram contrasting a prior art 64-bit electrical bus with the massively parallel 256-bit optical bus of the present invention.

Detailed Description of the Invention

Referring to **FIG. 1**, the apparatus is conceptually contrasted with prior art. A traditional 64-bit electrical bus (100) requires a large number of physical pins (101) on a CPU package and a corresponding set of 64 parallel electrical traces (102) on a circuit board. The massively parallel optical bus (110) of the present invention achieves a much wider 256-bit data path using a microscopic array of only 11 parallel optical channels (pixels) (111) that transmit the data word in a single cycle through an optical medium (112), such as an image conduit or free-space link. The physical footprint is drastically reduced, and the data throughput is increased fourfold, assuming an equivalent clock cycle.

Claims

What is claimed is:

1. A computer data bus apparatus for transmitting data words between components, comprising:
    - a plurality of parallel optical channels; and
    - a controller configured to cause a single data word with a bit-width greater than 128 bits to be transmitted in a single cycle by encoding portions of the data word across the plurality of parallel optical channels.
2. The apparatus of claim 1, wherein the bit-width of the data word is 256 bits.
3. The apparatus of claim 1, wherein the bit-width of the data word is 1024 bits.
4. The apparatus of claim 1, wherein each parallel optical channel is a pixel within a two-dimensional array of pixels.
5. The apparatus of claim 4, wherein a 256-bit data word is transmitted using 11 or fewer pixels.

6. The apparatus of claim 4, wherein a 1024-bit data word is transmitted using 43 or fewer pixels.

7. The apparatus of claim 1, wherein the bus is an implementation of the optical interface described in U.S. Patent Application No. 19/559,705.

Drawings

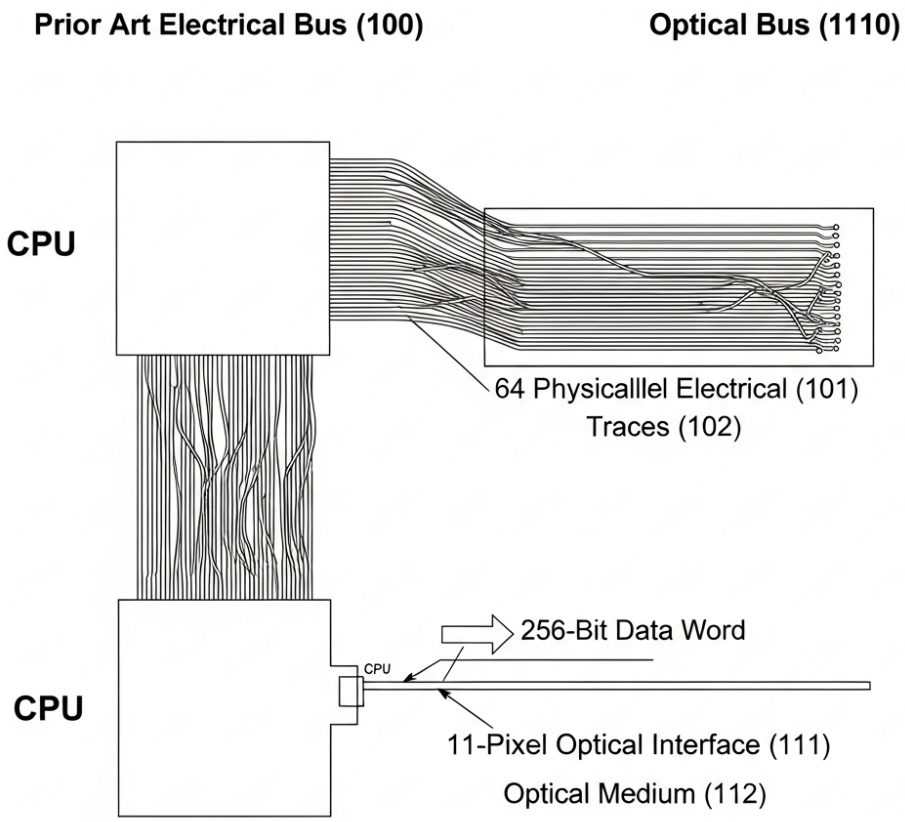


Figure 1: Conceptual diagram contrasting a prior art 64-bit electrical bus with the massively parallel 256-bit optical bus of the present invention.